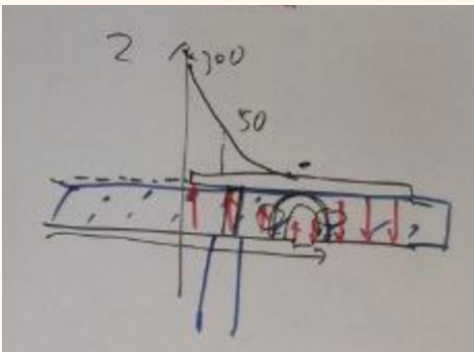


Step	Formula	Design
1. Width, W	$W = \frac{(\lambda/2)}{\sqrt{\frac{1+\epsilon_r}{2}}} \text{ (mm)}$	$W = 40.82 \text{ (mm)}$
2. Effective dielectric constant (for $\frac{h}{W} < 1$)	$\epsilon_{\text{reff}} = \frac{1+\epsilon_r}{2} + \frac{\epsilon_r-1}{2} \sqrt{1 + 12 \frac{h}{W}}$	$\epsilon_{\text{reff}} = 3.77$
3. Length, L	$L = \frac{(\lambda/2)}{\sqrt{\epsilon_{\text{reff}}}} - 0.824h \left[\frac{(\epsilon_{\text{reff}}+0.3)(\frac{W}{h}+0.264)}{(\epsilon_{\text{reff}}-0.258)(\frac{W}{h}+0.8)} \right] \text{ (mm)}$	$L = 30.03 \text{ (mm)}$
4. Width of stripline, W_s	for microstrip line characteristic impedance of $Z_o = Z_{in} = 50 \text{ } (\Omega)$	$W_s = 3.137 \text{ (mm)}$

Initial dimensions for a square patch: $L=W=30.03 \text{ mm}$.

- L determines resonant frequency and
- W affects impedance matching
- W must be shorter than half wavelength
- λ in free space
- notch to tune for input impedance



- change feeding type to feed with better input impedance

Feeding structures:

- Microstrip fed
- Probe fed
- Slot coupled,
- Edge coupled

